

Integrated Digitization & Clinical Decision Support Framework for Hemodialysis Units, Graduation Project Proposal

Prepared for: Dr. Mohamed Abdou, Office of the Dean

Prepared by:

- Aya Anwer, ID 202301817, CGPA 3.78
 - Ahmed Nassar, ID 202304446, CGPA 3.7245
 - Amr Khaled, ID 202301964, CGPA 3.47
 - Ahmed Gamal, ID 202304072, CGPA 3.45
 - Abdelrahman Seif, ID 202304069, CGPA 3.45
 - Shaza Hany, ID 202301617, CGPA 3.4
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2. Problem Statement

Chronic kidney disease is now the fifth leading cause of mortality in Egypt, with a dialysis population of roughly 54,000 to 59,000 patients generating an estimated 9.2 million hemodialysis sessions per year.^[1]

Patient safety in dialysis units is heavily time constrained. Session timing and medication timing are loosely directed by physicians managing large caseloads, and this creates communication gaps that directly affect patient vitals:

- Physicians occasionally prescribe mismatched medication or forget to update a prescription, simply due to patient volume.
- Nurses miss medication timing or vital sign documentation because of paperwork overhead, causing real data loss and reducing both nurse and patient visibility into the session.

- Dialysis machine settings are often generic rather than based on a patient's own history and lab trends, which nursing staff report directly contributes to fatigue and lower session quality.
- Because there is no unit wide aggregation of patient data, collective patterns, such as a shared mineral decline across multiple patients pointing to a water treatment or equipment fault, almost always go unnoticed until they cause acute harm.

2.1 Documented precedent for unit wide equipment failure

These are not hypothetical risks. In a documented outbreak at a London hemodialysis unit, a cracked water softener casing allowed calcium and magnesium to break through into the dialysate, exposing 30 patients to hard water syndrome: corrected serum calcium rose from a normal baseline to a toxic mean, and patients presented with acute hypertension, vomiting, and new onset arrhythmias that were initially misdiagnosed as unrelated conditions.^[6]

Separately, an outbreak at Guro Hospital in Seoul saw chloramine levels in treated water exceed safety limits due to municipal water contamination, triggering acute hemolytic anemia in 34 of 41 maintenance dialysis patients, an 83 percent attack rate, with hemoglobin levels collapsing and transfusion needs tripling.^[7]

Similar chloramine linked hemolytic outbreaks have been documented in Minas Gerais, Brazil, and a nationwide US recall of defective dialysis tubing caused mechanical hemolysis and multiple patient deaths across three states.^[8,9]

Each of these cases shares the same signature: a slow, easily monitorable water quality or equipment drift that produced a sudden, simultaneous decline across many patients at once, a pattern that paper based, unit isolated recordkeeping is structurally unable to catch before harm occurs.

2.2 Findings from our own field survey

During early field engagement with unit staff, two specific, previously unescalated incidents were confirmed that reinforce this problem statement directly:

- An episode in which an entire month's batch of patient lab results came back anomalously uniform, prompting patients to deny that the results belonged to them at all. This data integrity failure went unresolved because no historical, per patient baseline existed to flag the anomaly automatically.
- Confirmation, from a reviewer familiar with the unit's incident history, that chlorine and chloramine neglect in the unit's water purification system has occurred an estimated sixty times at this specific unit, with each occurrence going unrecorded and unescalated.

Both incidents point to the same root cause: the unit had no structured medication protocol and no simple, accessible patient history to catch the anomaly automatically. Both would have been readily detectable, and very likely preventable, under the system proposed here.

3. Existing Landscape

Two efforts are directly relevant and inform the scope of this proposal:

- FEDIS, Fixed Egypt for Digital Solutions, has already established a national dialysis machine automation system in Egypt. Our team has reached out to FEDIS directly. This framework is scoped to complement that infrastructure rather than duplicate it, by focusing on the clinical data analysis and decision support layer rather than machine automation itself.^[4]
- Egypt's Universal Health Insurance rollout is targeting national digital health coverage for 110 million citizens by 2032, which makes interoperability, HL7 and FHIR based data exchange, a design requirement rather than an afterthought.^[5]

Positioning this project as an addition to, rather than a replacement for, existing national infrastructure is a deliberate choice: it increases the likelihood of institutional support and avoids redundant engineering effort.

4. Proposed Solution

The framework is organized around four priorities, in order of clinical urgency.

4.1 Eliminate the paperwork blind spot, with IoT telemetry and patient support

Replace paper vitals, medication, and incident logs with structured digital entry at the point of care, so that no vital sign, medication timing, or adverse event is lost to clerical overhead.

- IoT telemetry capture: bedside and machine mounted sensors stream blood pressure, weight, and dialysate parameters directly into the patient record during the session, removing manual transcription as a point of failure rather than only digitizing the paper form that replaces it.
- Automated water quality telemetry: continuous electrical conductivity, pH, and chlorine or chloramine sensors placed at critical points in the water purification loop, replacing the single daily manual reading that historically leaves an average of 20 days a year with no data at all.^[10]
- Patient support setup: a lightweight patient facing interface or wearable link that surfaces the same session data to the patient, closing the loop described in Section 2 where patients had no way to independently confirm their own results.

Together these recover nursing time currently lost to paperwork and prevent the single most common and most preventable class of error: missed or mistimed documentation.

4.2 Digital, accessible dashboards for doctors and nurses

Doctors receive a clinical dashboard for reviewing patient trends, verifying and updating prescriptions, and communicating directly and securely with nursing staff. Nurses receive a counter mounted or mobile accessible interface showing live session status, medication schedules, and single tap incident reporting, replacing paper charts with something built for a time constrained clinical floor.

4.3 Data analysis module: per patient risk profile and per unit quality score

Every session record, pre and post vitals, machine settings, incidents, medication timing, and every periodic lab result, hemoglobin, iron studies, electrolytes, dialysis adequacy markers, feeds a longitudinal data model per patient. This enables two outputs:

- A per patient risk profile, tracking trends such as dry weight drift, blood pressure trajectory, and iron or hemoglobin trends that indicate oxygen needs during a session.
- A per unit quality score, aggregating patient outcomes across a unit to flag systemic issues, such as the unit wide mineral trend described in Section 2.1, before they cause harm.

4.4 Intelligent clinical decision support

Layer machine learning and clinical language models on top of the data foundation to support, not replace, physician judgement.

- Predictive intradialytic hypotension detection: intradialytic hypotension is the most common acute complication of hemodialysis, occurring in up to 40 percent of sessions. Validated models can predict onset 15 to 75 minutes in advance using session and vital trend data, including a recurrent neural network achieving an AUROC of 0.94, an XGBoost model achieving an AUROC of 0.936 using each patient's prior ten sessions as a personalized risk baseline, a CatBoost model achieving 0.845 to 0.880 across multiple case definitions, and a bidirectional LSTM achieving a sensitivity of 74.81 percent and specificity of 73.3 percent from live session data alone.^[12,13,14,15]
- Renal specific clinical decision support: a computerized order entry check that cross references a physician's prescription in real time against the patient's latest labs, estimated glomerular filtration rate, and dialysis schedule, the kind of check shown to reduce prescription errors by 81 to 86 percent in trials.^[16]
- MedGemma as a clinical security layer: a specialized medical language model, fine tuned on clinical guidelines, drug interaction data, and renal dosing standards, sits between the physician's order and the nursing station. It intercepts each order, checks it against the patient's real time renal function, electrolytes, and active medication list, and blocks and flags any contraindicated or unadjusted dose, such as unadjusted enoxaparin, before it ever reaches a nurse.

A comparable rule based decision support system for anemia management, deployed at a real hemodialysis center, raised physician compliance with computerized recommendations from 56.2 percent to 78.6 percent and measurably reduced anemia management failure rates, evidence that a well designed, low friction alert system changes clinical behavior rather than being ignored.^[17]

5. Data Foundation

The above modules are only possible with a well scoped data model. Two data categories are collected.

5.1 Per session records, collected by nurses every session

- Session identification: patient, session, date, and which machine and nurse handled it, so any issue can be traced back to a specific machine or shift.
- Pre session: weight on arrival, doctor set target or dry weight, fluid gained since last session, blood pressure, pulse, temperature.
- During session: planned versus actual fluid removal, machine dialysate settings, blood and dialysate flow rates, blood thinner dose, planned versus actual duration, in session blood pressure checks, and any incident, fainting, cramping, access issues, machine alarms, logged in a structured, searchable form rather than free text.
- Post session: weight, blood pressure, pulse, fluid actually removed, complications flag.
- Medication given: what, dose, time given, who administered it, and the gap between scheduled and actual time, directly targeting the miscommunication problem described in Section 2.

5.2 Monthly lab records, ordered by doctors

- Hemoglobin and hematocrit, for anemia and oxygen needs during sessions.
- Iron studies, ferritin and iron saturation, tracked over months.
- Potassium, sodium, calcium, phosphorus.
- Parathyroid hormone, albumin, creatinine, bicarbonate.
- Pre and post urea and a dialysis adequacy score.
- Inflammation marker, to catch problems before they become symptomatic.

5.3 Supporting records required for the analysis layer to function

- A dry weight history, not a single overwritable field, since the target itself changes over time.
- A prescription history that appends rather than overwrites, so an outdated prescription is never silently lost.
- A machine maintenance log, without which equipment linked pattern detection is not possible.
- A structured incident taxonomy, so nurse documented events are queryable rather than buried in prose.

5.4 Access and engagement already secured

- Permission has been obtained from a unit head physician to collect patient data and trial new solutions on the unit, independent of the dialysis machine settings themselves, for now.
- A collaborator with an existing dialysis focused application and data reach across multiple kidney units has joined the effort, with the ability to extend data collection to additional units if needed, and a connection to a dialysis focused health blogger who can help gather broader patient data, opinions, and trust.
- A contact at another hospital's dialysis unit, a physician also active in research, has agreed to support the project and has already flagged a specific technical opportunity: in session measurement of blood minerals as a potential life saving addition.
- Outreach is underway to the head of a university medical research institute to shape the data analysis component of the system with academic rigor.
- Reference is already available on the exact hemodialysis machine model in use in at least one partner unit, Fresenius 4008 S, which will inform integration requirements.

6. What We Are Requesting

Our primary request is for the Dean's support in two specific areas:

- Technical direction and institutional connections needed to implement and test the proposed telemetry and software on a live dialysis unit, including any ethics review or data governance guidance required for a pilot involving real patient data.
- Guidance and support in applying to a national accelerator or incubator program, so the project can be positioned for national scale coverage rather than remaining a single unit pilot.

We believe this project directly serves patient safety at a national scale while remaining realistic in scope: building the clinical data and decision support layer that complements, rather than duplicates, Egypt's existing national dialysis automation efforts.

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